



AI Cop?

Scientists have developed an experimental drone security system that can detect violent individuals in a crowd by using AI. <https://digit.in/jul18-75-1>



Kidney cancer cells play pretend

A recent research showed that kidney cancer cells travel around the human body by pretending to be white blood cells. <https://digit.in/jul18-75-2>

and an edge to edge mirror assembly. Each of the payloads would be launched independently, as and when the opportunity arises, piggybacking as secondary payloads of the main launches. Once in space, each hexagonal module would use a solar sail to navigate to the Sun-Earth Lagrange point 2 (L2). The solar sails also double up as a sunshield during assembly, protecting the sensitive instruments within the hexagons from being damaged by solar radiation. At L2, the telescope will assemble itself, without any additional intervention from humans. The combined gravity from the Earth and the Sun is stable in this location, which is where the space telescope can be parked. The Sun-Earth L2 point is the chosen destination for space telescopes, and hence, it is where the JWST will be deployed as well.

The concept is targeted for the Large Ultraviolet/Optical/Infrared Surveyor, a mission on the NASA roadmap for the 2020s. The Phase I of the study has been greenlit, where the details of how the components will travel to the target location will be worked out through simulations. This will in turn be used to figure out the required surface area of the solar sails to realise the mission. The exact mechanism of assembling the mirrors will also be validated. If Phase I of the study is successful, the team can apply for Phase II funding. This study is for a space telescope with an ap-



The AAReST telescope

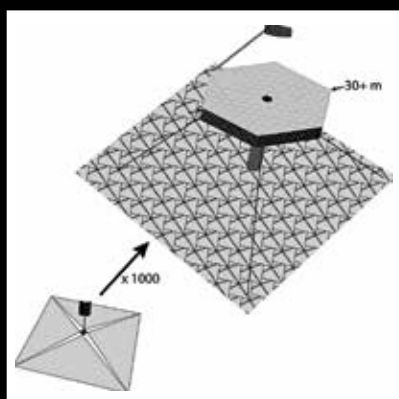
erture of 30 meters. However, there are even bigger telescopes being conceptualised, which would also require autonomous assembly in space.

Researchers from the University of Surrey and Caltech have proposed a telescope with an aperture of 100 meters. The assembly, is however more complicated. Apart from the modules, there would be additional external intervention in the form of a robot dedicated to the assembly process. The idea is to launch 300 support truss modules. These will provide a frame to support the main telescope. A six armed hexbot will first assemble the frame, and then crawl across it in the process of assembling the mirrors. Like a mountain climber, the robot would use three arms to anchor itself to the frame, while moving only one arm at a time to move. The two additional arms will be dedicated to the assembly process. The power for the robot will be drawn from the same solar panels used to power the telescope itself. The main mirror is made up of very thin, lightweight material. The materials have the ability to deform, to fit the shape of the aperture. The rest of the space observatory would be stretched out in a staggered manner. 400 meters from the main mirror, would be an optics and instrumentation unit. 400 meters beyond that would be a control unit. At this location, a 20 meter wide sun

shield would provide protection to the components of the observatory, and keep the instruments from overheating in outer space.

The Autonomous Assembly of a Reconfigurable Space Telescope (AAReST) mission, is a collaboration between Caltech, the University of Surrey, NASA/JPL and the Indian Institute of Space Science. It is a technology demonstration mission, that will show that in principle, space telescopes can be assembled in orbit. The components are a number of small nanosatellites known as CubeSats. The navigation systems of the various components use the COTS PrimeSense LIDAR, which is also used in the Microsoft Kinect. The design consists of a main mirror component, and six satellite mirrors that can autonomously undock and redock to provide various configurations to the telescope.

Self assembling space telescopes are still in the nascent stage, but will provide us with unprecedented access to the skies. These kinds of telescopes will fundamentally change how humanity looks at space. The telescopes will also be useful for dark matter and dark energy research. Telescopes with such large apertures mean that we will be able to directly image the surface details of exoplanets, something that is beyond the capabilities of even the JWST. 



The Savransky Telescope, with one module next to the fully assembled telescope with 100 modules